In [2]:
```
# import the libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

In [3]:
```
# Create the data
data = {
    'pages': [5, 12, 3, 8, 15, 4, 10, 6, 20, 7,
              9, 2, 14, 5, 11, 3, 8, 18, 6, 13,
              4, 16, 7, 10, 2, 12, 5, 9, 15, 6],

    'deadline_days': [14, 7, 21, 5, 10, 18, 6, 12, 8, 15,
                      4, 25, 9, 11, 7, 20, 6, 5, 14, 8,
                      16, 6, 10, 4, 22, 7, 13, 5, 9, 11],

    'rate_inr': [8000, 22000, 5000, 18000, 28000, 6500, 19000, 10000, 35000, 110
                 20000, 3500, 25000, 9000, 21000, 5500, 17000, 38000, 9500, 2400
                 7000, 32000, 13000, 23000, 4000, 22500, 8500, 19500, 29000, 115
}
```

In [4]:
```
df = pd.DataFrame(data)
```

In [6]:
```
df.head()
```

Out[6]:

	pages	deadline_days	rate_inr
0	5	14	8000
1	12	7	22000
2	3	21	5000
3	8	5	18000
4	15	10	28000

In [8]:
```
df.dtypes
```

Out[8]:
```
pages            int64
deadline_days    int64
rate_inr         int64
dtype: object
```

In [10]:
```
df.describe()
```

Out[10]:

	pages	deadline_days	rate_inr
count	30.000000	30.000000	30.000000
mean	8.833333	10.933333	16833.333333
std	4.885470	5.777085	9715.197243
min	2.000000	4.000000	3500.000000
25%	5.000000	6.250000	8625.000000
50%	8.000000	9.500000	17500.000000
75%	12.000000	14.000000	22875.000000
max	20.000000	25.000000	38000.000000

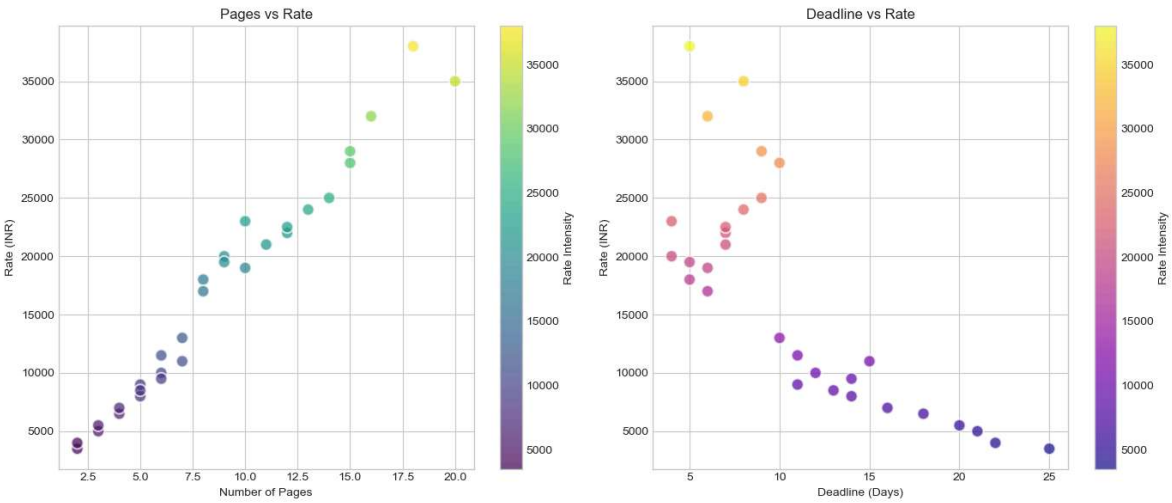
In [21]:

```
#Visualize the Data
plt.style.use('seaborn-v0_8-whitegrid')
fig, axis = plt.subplots(1, 2, figsize=(14, 6))

scatter1 = axis[0].scatter(df['pages'], df['rate_inr'],
                           c=df['rate_inr'], cmap='viridis', s=100, alpha=0.7, e
axis[0].set_xlabel('Number of Pages')
axis[0].set_ylabel('Rate (INR)')
axis[0].set_title('Pages vs Rate')
plt.colorbar(scatter1, ax=axis[0], label='Rate Intensity')

scatter2 = axis[1].scatter(df['deadline_days'], df['rate_inr'],
                           c=df['rate_inr'], cmap='plasma', s=100, alpha=0.7, ed
axis[1].set_xlabel('Deadline (Days)')
axis[1].set_ylabel('Rate (INR)')
axis[1].set_title('Deadline vs Rate')
plt.colorbar(scatter2, ax=axis[1], label='Rate Intensity')

plt.tight_layout()
plt.show()
```



In [27]:

```
# Prepare data for training
X = df[['pages','deadline_days']]
y = df['rate_inr']
```

In [28]:
```
# split the data into training and testing sets
random_state = 42
X_train,X_test,y_train,y_test = train_test_split(X,y, test_size =0.2)
```

In [29]:
```
# create and train the model
model = LinearRegression()
model.fit(X_train,y_train)
```

Out[29]:
```
▾ LinearRegression
▶ Parameters
```

In [35]:
```
# check model parameters
pages_coeff = model.coef_[0].round()
deadline_coeff = model.coef_[1].round()
intercept = model.intercept_
```

In [36]:
```
pages_coeff, deadline_coeff, intercept
```

Out[36]:
```
(np.float64(1786.0), np.float64(-207.0), np.float64(3117.117197413012))
```

In [38]:
```
# Make predictions on test data set
y_pred = model.predict(X_test)
```

In [39]:
```
y_pred
```

Out[39]:
```
array([21313.8212278 ,  2129.86580498, 16369.92474319, 20149.44084059,
        9353.5206543 , 28043.8503809 ])
```

In [40]:
```
results = pd.DataFrame({'Actual':y_test.values, 'Predicted': y_pred.round()})
results
```

Out[40]:

	Actual	Predicted
0	21000	21314.0
1	4000	2130.0
2	18000	16370.0
3	23000	20149.0
4	8500	9354.0
5	29000	28044.0

In [43]:
```
# Model accuracy
from sklearn.metrics import r2_score
score = r2_score(y_test, y_pred)
score
```

Out[43]:
```
0.9634109320332438
```

In [49]:
```
# Answer for given question
new_project = [[10,5]]
predicted_rate = model.predict(new_project)
```

```
C:\Users\G.Guna\desktop\ML\heart_disease_project\env\Lib\site-packages\sklearn\ut
ils\validation.py:2749: UserWarning: X does not have valid feature names, but Lin
earRegression was fitted with feature names
  warnings.warn(
```

In [50]:
```
print(f"\nQuestion: 10-page website, 5-day deadline")
print(f"Predicted Recommended Rate: ₹{predicted_rate[0]:,.0f}")
```

```
Question: 10-page website, 5-day deadline
Predicted Recommended Rate: ₹19,942
```

In [ ]: